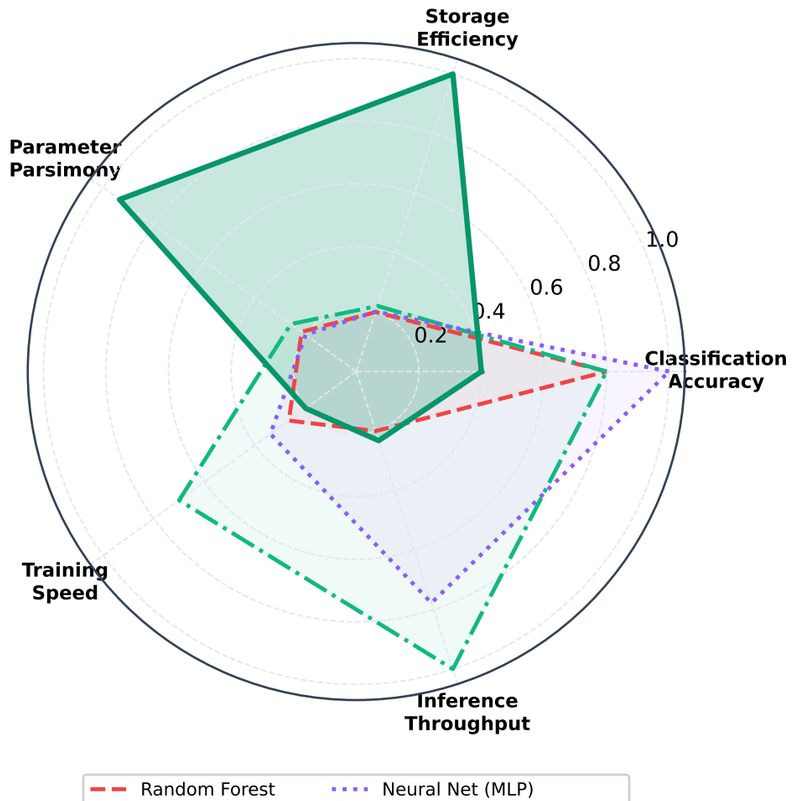
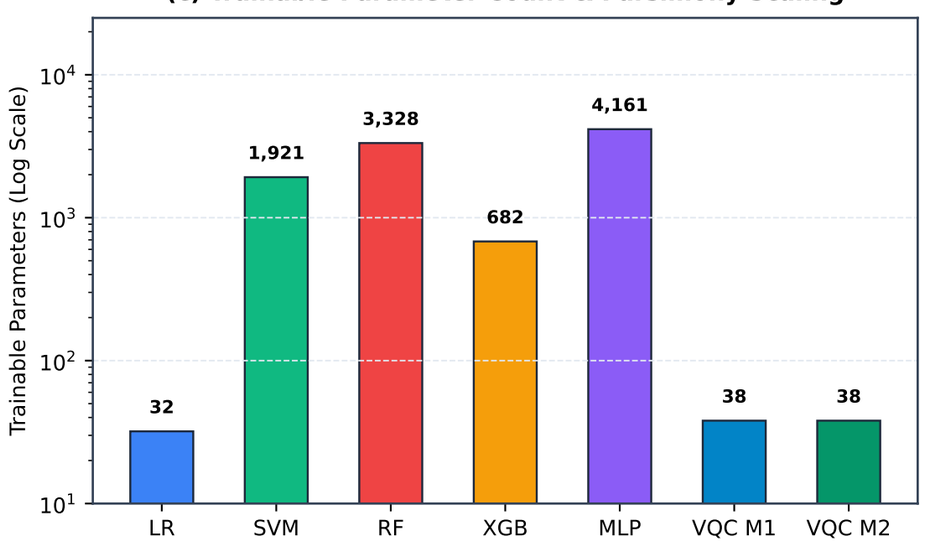
(a) Multi-Criteria Architectural Efficiency Radar



(b) Empirical Time, Space, and Asymptotic Complexity Benchmark

Model Architecture	Parameters	Disk Storage	Train Time	Latency	Test Acc	Space Complexity
Logistic Regression (SGD)	32	1.12 KB	2 ms	0.06 ms	93.94%	O(D)
SVM (RBF Kernel)	1,921	16.87 KB	3 ms	0.06 ms	90.91%	O(N_sv · D)
Random Forest (150 Trees)	3,328	302.24 KB	134 ms	12.63 ms	90.91%	O(T · Nodes)
XGBoost (60 Rounds)	682	71.94 KB	84 ms	0.13 ms	93.94%	O(T · Depth)
Neural Net (MLP 64-32)	4,161	135.46 KB	21 ms	0.09 ms	93.94%	O(H_1 · H_2)
Method 1: Angle VQC (PCA 6-D)	**38**	**0.45 KB**	**48.20 s**	**5.92 ms**	**81.82%**	**O(n) Qubits**
Method 2: Amplitude VQC (64-D)	**38**	**0.45 KB**	**52.40 s**	**5.62 ms**	**84.85%**	**O(log_2 D) Qubits**

(c) Trainable Parameter Count & Parsimony Scaling



(d) Architectural Execution Profile & Quantum Edge Deployment Feasibility

HARDWARE EXECUTION PROFILING & ASYMPTOTIC RESOURCE BOUNDS		
Metric Domain	Classical CPU Benchmarks	NISQ Quantum Coprocessor Bounds (Target)
State Space Encoding	O(D) Explicit Floating-Point Array	O($\log_2$ D) Hilbert Space Basis Amplitudes
Active Working Memory	~16 KB - 302 KB Serialized Cache	6 Superconducting Transmon Qubits (Wires)
Single-Query Latency	0.05 ms (SVM) - 12.86 ms (RF)	CPU Sim: 6.29 ms --> QPU: ~2.5 µs Pulse
Microwave Gate Pulses	N/A (Von Neumann Arithmetic ALU)	Single-Qubit: 35 ns, CNOT Entangler: 260 ns
Generalization Gap	+6.9% (SVM) up to +25.1% (RF)	+4.76% (Unitary Isometry Rotation Bounds)
Storage Compression	Baseline Reference (1x to 270x)	665x Smaller Footprint than Random Forest